Report 3

Breast-cancer-prediction

Dataset = WISCONSIN DIAGNOSIS BREAST CANCER(WDBC)

Friday, 05th july 2024

Content:

Comparison of Feature Extraction Approaches for Cancer Diagnosis and Prognosis

Information:

- The advanced report provides a detailed comparison of the feature extraction approaches used in two recent studies: "[1]Nuclear Feature Extraction for Breast Tumor Diagnosis[1]" and "CT Texture Features Are Associated with Overall Survival in Pancreatic Ductal Adenocarcinoma[2]."

[1] The "Nuclear Feature Extraction for Breast Tumor Diagnosis" study is the fundamental article that has led to the development of the widely used WDBC (Wisconsin Diagnostic Breast Cancer) dataset.

Introduction :

Accurate and comprehensive characterization of cancer is crucial for effective diagnosis, treatment, and prognosis. Two recent studies have explored different approaches to feature extraction from medical imaging data, each focusing on a specific cancer type and clinical application. The first study, "Nuclear Feature Extraction for Breast Tumor Diagnosis," examined cellular-level features from microscopic images, while the second study, "CT Texture Features Are Associated with Overall Survival in Pancreatic Ductal Adenocarcinoma," investigated tissue-level texture features from contrast-enhanced CT scans. This report provides a detailed comparison of the feature extraction methods.

Table 1: Comparison between these two articles.

Characteristic	**Nuclear Feature Extraction for Breast Tumor Diagnosis**	**CT Texture Features are Associated with Overall Survival in Pancreatic Ductal Adenocarcinoma**
Tumor Type	Breast Tumors	Pancreatic Ductal Adenocarcinoma (PDAC)
Imaging Modality	Fine Needle Aspirate (FNA) Microscopy Images	Contrast-Enhanced Computed Tomography (CT)
Feature Type	Nuclear Morphology(Nuclear size, Nuclear shape, Nuclear texture)	Texture Features(Gray-level co-occurrence matrix (GLCM) features, Histogram-based features, Wavelet-based features)
Features Extracted	10 Nuclear Features (size, shape, texture)	5 CT Texture Features (uniformity, entropy, dissimilarity, correlation, inverse difference normalized)
Feature Selection	Linear Programming-based Classifier (MSM-Tree) to identify 3 best features (mean	Receiver Operating Characteristic (ROC) analysis, Cox Regression,

	texture, worst area, worst smoothness)	Kaplan-Meier to identify dissimilarity and inverse difference normalized as prognostic features
Image Processing	Interactive Segmentation	Automated Segmentation
Field of View	Focused on individual cells	Whole tumor and surrounding tissues
Radiation Exposure	None	Ionizing radiation (X-rays)
Potential for Extracting Texture Features	The microscopic images of cells could potentially be used to extract texture features, such as: - Gray-level co-occurrence matrix (GLCM) features - Histogram-based features - Wavelet-based features This would require additional image processing and analysis beyond the nuclear morphology features.	The CT images are already well-suited for extracting texture features, as demonstrated in the study.
Clinical Applications	Diagnostic cytology, cancer grading	Tumor staging, treatment planning, response assessment

Conclusion:

1. **Imaging Modality Mismatch:** The "CT Texture Features" study utilized contrast-enhanced CT images, which capture tissue-level characteristics of the tumor. In contrast, the "Nuclear Feature Extraction" study focused on microscopic images of individual cells obtained through fine needle aspiration. The texture analysis techniques developed for CT images may not be directly applicable to the cellular-level microscopic images.
2. **Anatomical Scale Difference:** The texture features extracted from the CT images in the "CT Texture Features" study represent the tissue-level characteristics of the tumor. However, the microscopic images in the "Nuclear Feature Extraction" study capture cellular-level details, which require different image processing and feature extraction approaches compared to the tissue-level texture analysis.

Refrences :

1. Street, W.N., Wolberg, W.H. and Mangasarian, O.L., 1993, July. Nuclear feature extraction for breast tumor diagnosis. In *Biomedical image processing and biomedical visualization* (Vol. 1905, pp. 861-870). SPIE.https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=Nuclear+Feature+Extraction+for+Breast+Tumor+Diagnosis&btnG=
2. Eilaghi, A., Baig, S., Zhang, Y., Zhang, J., Karanicolas, P., Gallinger, S., Khalvati, F. and Haider, M.A., 2017. CT texture features are associated with overall survival in pancreatic ductal adenocarcinoma–a quantitative analysis. BMC medical imaging, 17, pp.1-7.https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=CT+Texture+Features+Are+Associated+with+Overall+Survival+in+Pancreatic+Ductal+Adenocarcinoma&btnG=